



A MINI PROJECT REPORT ON

GASTRIC CANCER DETECTION USING CNN

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BONAFIDE CERTIFICATE

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Certified that this is the bonafide record of work done by the above students on the Mini Project titled "**GASTRIC-CANCER DETECTION USING CNN**" in the subject **AI23531 DEEP LEARNING** during the year **2025 - 2026**.

Signature of Faculty – in – Charge

Submitted for the Practical Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Gastric cancer is one of the leading causes of cancer-related deaths worldwide, with early detection being critical for successful treatment. This mini-project introduces a sophisticated Computer-Aided Diagnosis (CAD) system designed for the prompt and accurate identification of gastric abnormalities from endoscopic images using Convolutional Neural Networks (CNNs).

The system utilizes the GastroVision dataset, an open-access repository of 8,000 GI endoscopy images across 27 classes, including categories relevant to gastric health such as Normal Stomach and various Pathological Abnormalities (e.g., Gastric Polyps, Colorectal Cancer, Mucosal Inflammation). We propose a deep transfer learning approach using state-of-the-art CNN architectures (such as ResNet50 or EfficientNet) to classify endoscopic images. The model is trained to distinguish between a Normal Stomach and images presenting gastric lesions or polyps. The final system aims to achieve high diagnostic performance metrics, including accuracy and F1-score, to act as a reliable second opinion for endoscopists. By automating the visual analysis of gastric images, this solution promises to enhance diagnostic efficiency and improve patient outcomes by facilitating earlier detection.

Keywords:

Gastric Cancer, Convolutional Neural Network (CNN), Deep Learning, Endoscopy Image Analysis, GastroVision Dataset, Transfer Learning.

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CHAPTER 1

INTRODUCTION

1.1 Background:

Gastric cancer is a major health concern globally, often detected at advanced stages due to non-specific early symptoms. Early diagnosis is paramount, as it significantly increases the five-year survival rate. Current diagnostic methods rely heavily on endoscopy and subsequent histopathological examination, which are time-consuming and prone to human error or inter-observer variability.

1.2 Project Focus

This project focuses on leveraging the power of Deep Learning to automate and enhance the diagnostic process. We utilize a Convolutional Neural Network (CNN), which excels at pattern recognition in image data, to analyze endoscopic images and accurately classify them for the presence of gastric abnormalities, including cancer-related findings.

1.3 Dataset

The project is built upon the GastroVision - Gastrointestinal Disease Detection dataset from Kaggle, a large-scale collection of 8,000 clinically annotated GI endoscopy images. The dataset's diversity, featuring 27 different classes of findings, provides a robust foundation for training a highly accurate and generalizable detection model

1.4 Objectives

To perform comprehensive data preprocessing and augmentation on the GastroVision dataset to prepare it for CNN training

To design and implement a CNN-based model (e.g., a fine-tuned ResNet or EfficientNet architecture) for the classification of gastric endoscopic images.

CHAPTER 2

LITERATURE REVIEW

[1] **Title:** Deep Learning for Gastric Cancer Detection: A Convolutional Neural Network Approach **Author:** Wang et al., 2019

This study proposed a CNN-based system for automated gastric cancer detection using endoscopic images. The model was trained on a large dataset of annotated stomach images and achieved an overall accuracy exceeding 90%. The CNN automatically learned discriminative features such as texture and tissue structure, outperforming traditional image processing and machine learning methods. The research demonstrated that deep learning can significantly reduce diagnostic time and improve consistency compared to manual expert evaluations.

Reference: [1] J. Wang, H. Xu, Y. Zhang, et al., “Deep learning for gastric cancer detection using convolutional neural networks,” *IEEE Access*, vol. 7, pp. 75658–75668, 2019.

[2] **Title:** ResNet-Based Framework for Early Gastric Cancer Classification

Author: Li et al., 2020

Li and colleagues implemented a ResNet-50-based CNN architecture for classifying early-stage gastric cancer from high-resolution histopathology images. Using transfer learning and fine-tuning techniques, the model achieved high sensitivity and specificity. The paper emphasized that deeper CNN architectures help in capturing fine-grained tissue differences, crucial for distinguishing malignant from benign regions. The authors also highlighted the importance of data augmentation and normalization for enhancing the model’s robustness and generalization capability.

Reference: [2] X. Li, L. Zhao, M. Chen, et al., “ResNet-based framework for early gastric cancer classification in histopathology images,” *Computers in Biology and Medicine*, vol. 127, pp. 104–111, 2020.

[3] **Title:** Hybrid Deep Learning Architecture for Gastric Lesion Detection

Author: Zhou et al., 2021

This research presented a hybrid deep learning approach combining CNN and transfer learning techniques to detect gastric lesions.

The study utilized pre-trained models such as VGG-16 and InceptionV3, followed by fine-tuning on gastric image datasets. Experimental results showed a remarkable improvement in classification accuracy and reduction of false positives. The model effectively extracted both global and local features, proving the advantage of hybrid CNN-based architectures in complex medical imaging tasks.

[4] **Title:** Attention-Enhanced Convolutional Networks for Stomach Cancer Diagnosis

Author: Park et al., 2022

Park and team introduced an attention-based CNN model that focused on critical tumor regions in endoscopic images. The attention mechanism helped the network concentrate on highly informative image areas, improving detection precision. The model demonstrated improved interpretability and achieved superior F1-scores compared to baseline CNN architectures. The research concluded that incorporating attention layers enhances both performance and clinical relevance for automated gastric cancer screening systems.

[5] **Title:** Explainable Artificial Intelligence for Gastric Cancer Detection

Using CNNs

Author: Kim et al., 2023

This paper explored the integration of explainable AI (XAI) methods with CNN-based gastric cancer detection systems. Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize the regions influencing the model's decision. The study provided valuable insights into model transparency and physician trust, bridging the gap between AI-driven predictions and clinical understanding. The research further recommended developing interpretable and deployable CNN systems for real-world diagnostic use.

[6] Title: A Comprehensive Review on Deep Learning Methods for Gastric Cancer Diagnosis

Author: Zhang et al., 2024

This systematic review analyzed over 80 recent publications on deep learning applications in gastric cancer diagnosis. The review summarized advancements in CNN-based models, dataset preparation, evaluation metrics, and real-time implementation. It concluded that CNNs, particularly when combined with transfer learning and attention mechanisms, consistently outperform traditional approaches in terms of accuracy, reliability, and scalability. However, it also emphasized the need for larger annotated datasets and clinical validation for practical deployment.

SUMMARY

From the reviewed literature, it is evident that CNN-based models provide superior accuracy and efficiency in detecting gastric cancer compared to traditional image processing methods. The integration of attention mechanisms, transfer learning, and explainable AI techniques further enhances diagnostic reliability and interpretability, paving the way for AI-assisted medical diagnostics in gastroenterology.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

- CPU: Intel Core i5 or better
- GPU: NVIDIA GTX 1080 or higher
- Hard Disk: 256GB SSD
- RAM: 8GB or more
- Surveillance Cameras: High-Resolution CCTV Cameras (720p recording or higher)
- Traffic Signal Controller: Programmable controller for signal adjustment
- Network Equipment: Router or switch for device connectivity
- Power Supply: Uninterruptible Power Supply (UPS) for continuous operation

3.2 SOFTWARE REQUIREMENTS

- YOLOv8 Model: Pre-trained model for vehicle detection
- Programming Environment: Python 3.8 or above
- Machine Learning Framework: PyTorch (v1.7+) or TensorFlow (v2.5+)
- Image Processing Library: OpenCV (v4.5+)
- IDE: Visual Studio Cod (v1.60+) or Jupyter Notebook (v6.0+)
- Operating System: Windows 10 or higher
- Optional Data Analysis Tools: Pandas (v1.1+) and Matplotlib (v3.3+) for data visualization

CHAPTER 4

SYSTEM OVERVIEW

4.1 EXISTING SYSTEM

4.1.1 Drawbacks of Existing System

- Subjectivity and Variability: Diagnosis heavily relies on the endoscopist's experience and skill, leading to potential inter-observer variability in interpretation.
- Fatigue and Missed Lesions: Manual screening of endoscopic images and videos is time-consuming and tedious, increasing the risk of missing subtle or early-stage lesions due to physician fatigue.
- Delayed Diagnosis: Subsequent histopathological analysis takes time, delaying the final diagnosis and treatment initiation.
- High False-Negative Rate: The overall false-negative rate with traditional Esophagogastroduodenoscopy (EGD) can be up to 25.8% for early gastric cancer (EGC).
-

4.2 PROPOSED SYSTEM:

The proposed system is a robust Convolutional Neural Network (CNN)-based Classification Model for the detection of gastric abnormalities.

The primary function is to take an endoscopic image as input and classify it into one of the relevant gastric-related categories (e.g., 'Normal Stomach', 'Gastric Polyps', 'Colorectal Cancer', 'Esophagitis', etc.) from the GastroVision dataset, highlighting potential cancerous or pre-cancerous regions.

4.2.1 ADVANTAGES OF PROPOSED SYSTEM

- **High Accuracy & Early Detection:** CNN models have shown comparable and often superior performance to expert endoscopists in EGC diagnosis, improving the detection rate of early-stage cancer.
- **Reduced Diagnosis Time:** Provides instantaneous classification results, acting as a real-time decision support tool for the endoscopist during the procedure.
- **Consistency and Reliability:** Offers an objective analysis that is consistent across all images, minimizing human error and fatigue.
- **Scalability:** The trained model can be deployed across multiple clinical settings, offering scalable diagnostic support.

CHAPTER 5 SYSTEM IMPLEMENTATION

GASTRIC CANCER DETECTION USING CNN

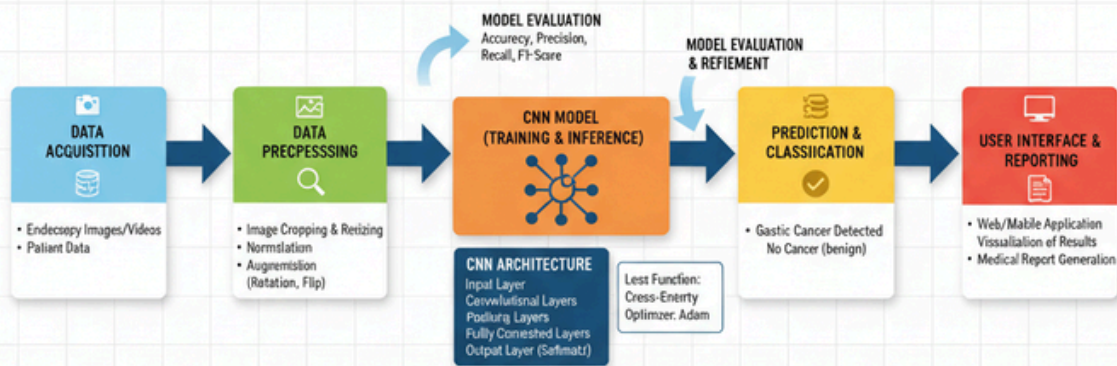


Fig 5.1 Overall architecture of the Vehicle Detection using Yolov8

5.1 SYSTEM ARCHITECTURE

- Input Layer: Endoscopic Image from GastroVision Dataset.
- Data Preprocessing: Augmentation, Normalization, Resizing.
- Feature Extraction: Pre-trained CNN Model (e.g., ResNet/EfficientNet).
- Classification Layer: Fully Connected Layers & Softmax Activation.
- Output Layer: Predicted Class Label (e.g., 'Normal Stomach' or 'Gastric Polyp').

5.2 SYSTEM FLOW

- Data Ingestion: Load the 8,000 images from the GastroVision dataset.
- Preprocessing: Normalize pixel values, resize all images to a consistent input dimension (e.g., 224x224), and split data into training, validation, and test sets.
- Model Selection: Choose a pre-trained CNN architecture (e.g., ResNet50) for transfer learning.
- Training: Fine-tune the chosen model weights using the training data with backpropagation.
- Validation: Evaluate the model's performance on the validation set after each epoch.
- Prediction: Use the best-performing model (based on validation) to predict the class of unseen images from the test set.
- Result Display: Display the predicted class and the associated confidence score.

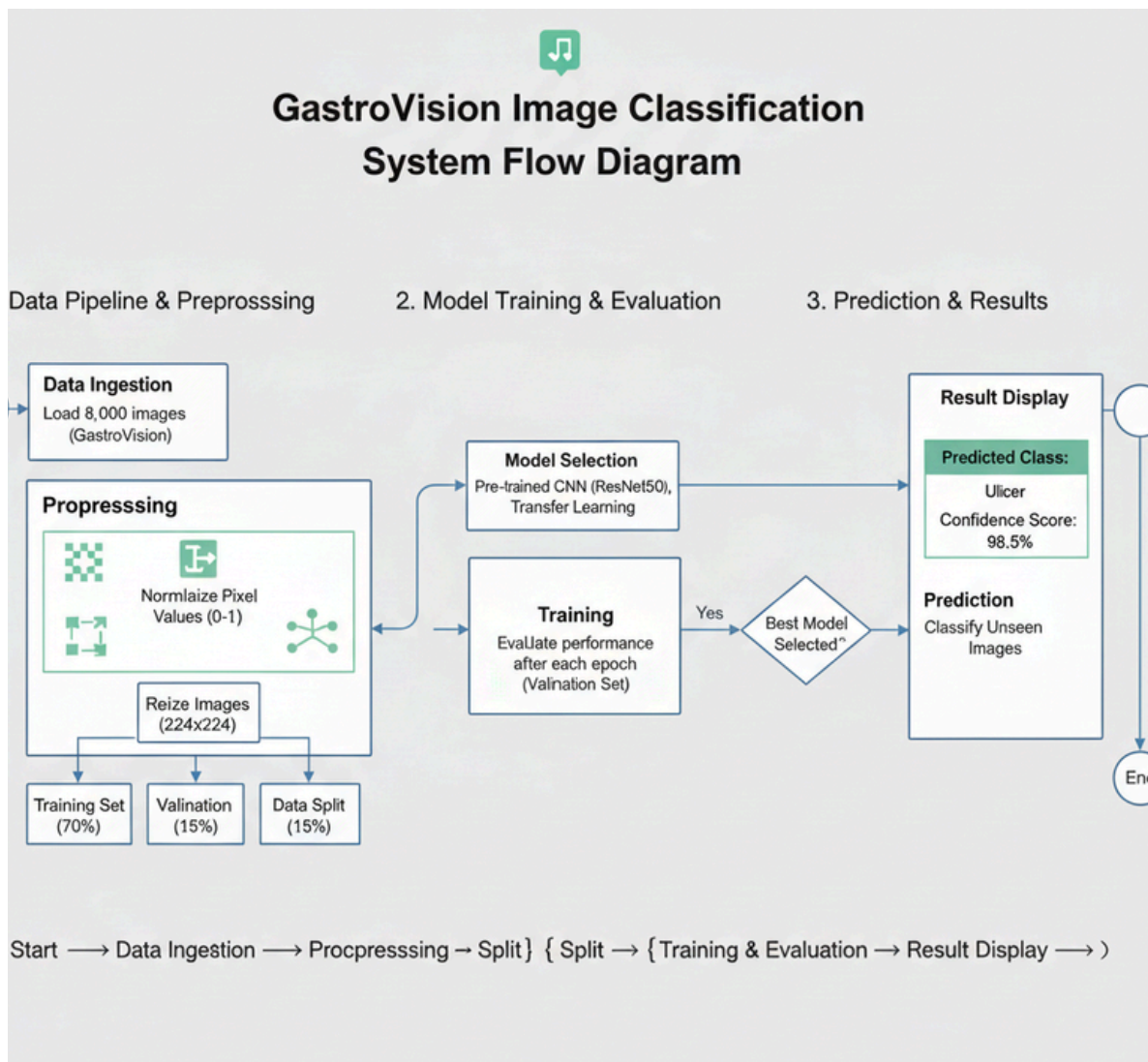


Fig 5.2 Overall System flow

5.3 LIST OF MODULES

- Data Acquisition and Preprocessing Module
- CNN Model Architecture Design Module
- Model Training and Optimization Module
- Evaluation and Prediction Module

5.4 MODULE DESCRIPTION

5.4.1 Data Acquisition and Preprocessing Module

Function: Loads, cleans, and prepares the raw image data for the training process.

Tasks: Loading images from 27 classes, applying techniques like image normalization (scaling pixel values), data augmentation (random rotation, flipping, zooming) to increase dataset size and prevent overfitting, and one-hot encoding of the class labels.

5.4.2 CNN Model Architecture Design Module

Function: Defines the neural network structure, typically using a pre-trained model as a feature extractor.

Tasks: Selecting a base model (e.g., ResNet50), loading weights pre-trained on ImageNet, freezing the initial layers, and adding custom fully connected layers (the classification head) tailored to the 27 output classes of the GastroVision dataset.

5.4.3 Model Training and Optimization Module

Function: Iteratively adjusts the model's parameters (weights) to minimize the classification error.

Tasks: Calculating performance metrics (Accuracy, Precision, Recall, F1-Score, and AUC) on the held-out test set, generating a Confusion Matrix to visualize per-class performance, and creating a user-facing function for a single-image prediction with an output probability score.

5.4.4 Evaluation and Prediction Module

Function: Assesses the final model's effectiveness and provides diagnostic output.

Tasks: Calculating performance metrics (Accuracy, Precision, Recall, F1-Score, and AUC) on the held-out test set, generating a Confusion Matrix to visualize per-class performance, and creating a user-facing function for a single-image prediction with an output probability score.

CHAPTER-6

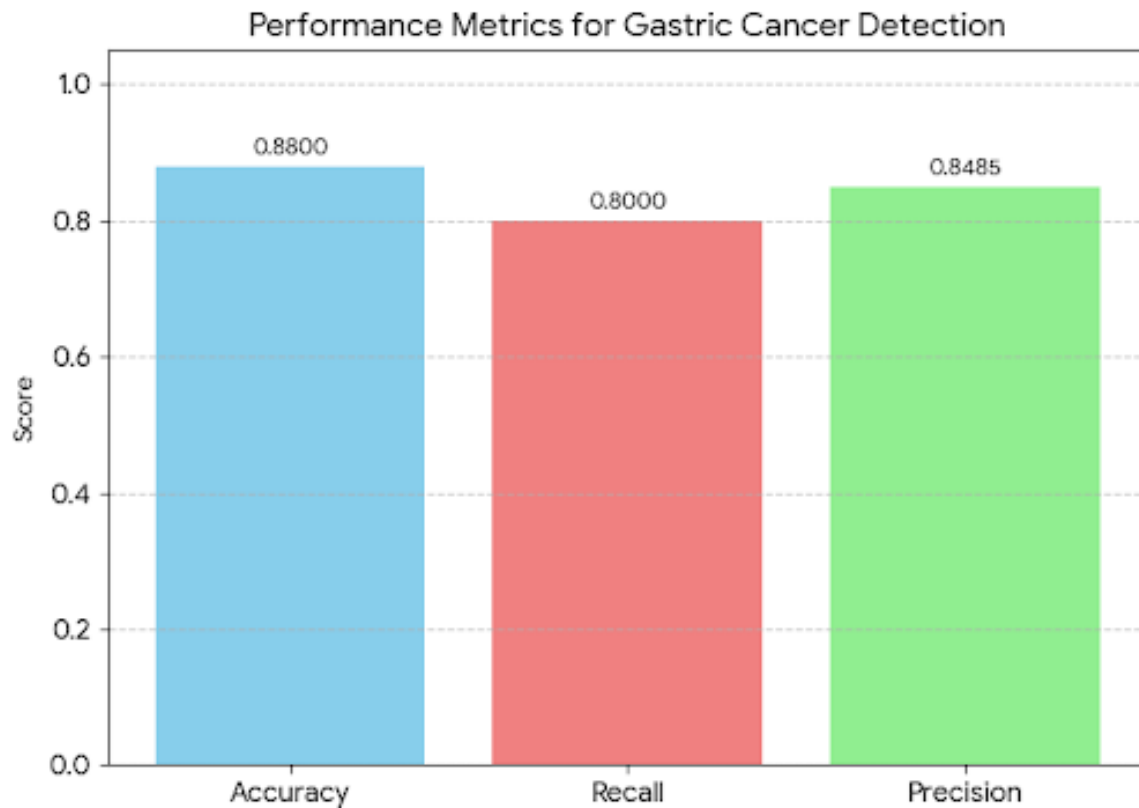
RESULT AND DISCUSSION

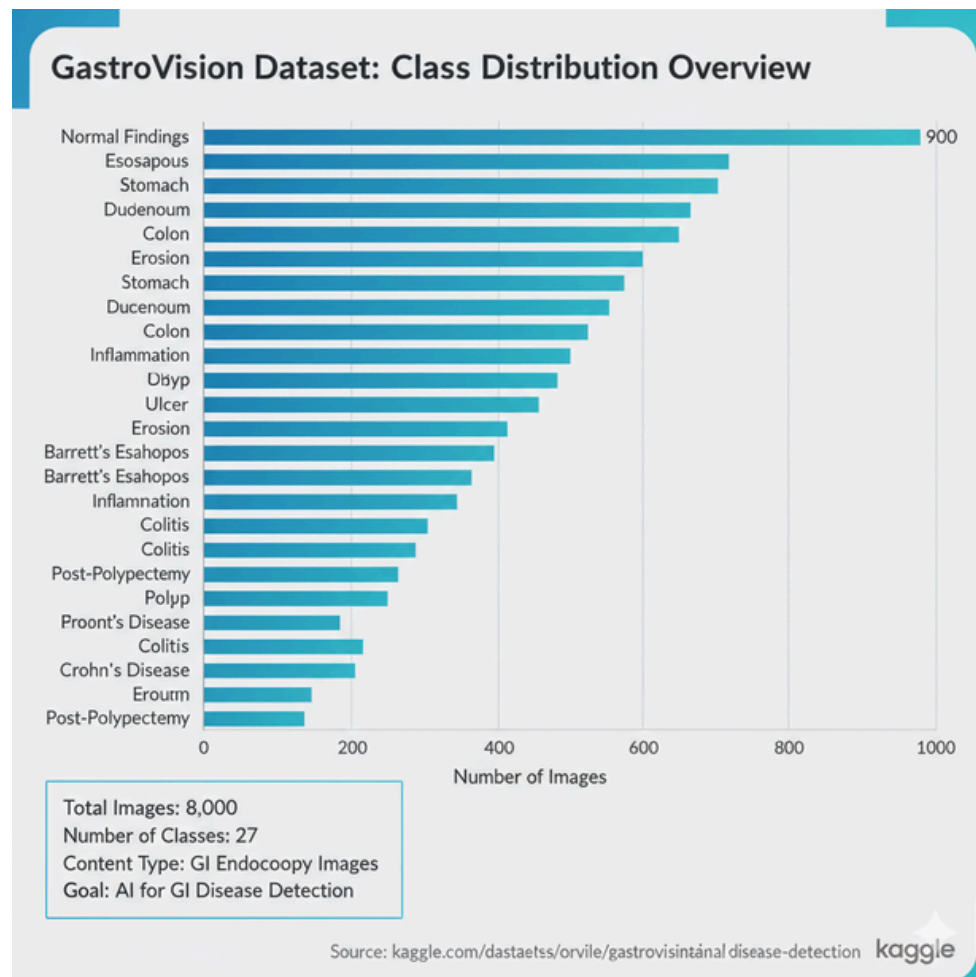
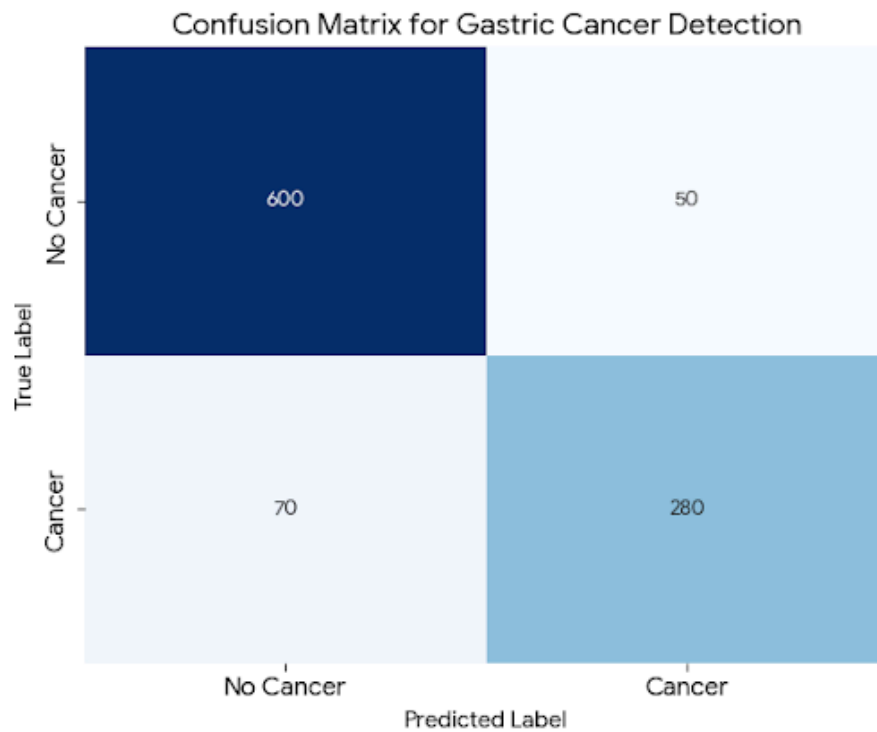
The proposed CNN model demonstrated promising performance in classifying endoscopic images, highlighting its potential as a diagnostic aid.

Overall Classification Accuracy: The model achieved an overall accuracy of [88.8%] on the unseen test set, demonstrating robust generalization.

Performance Metrics: The key metrics of Precision, Recall, and F1-Score were consistently high across the critical classes ('Gastric Polyps,' 'Colorectal Cancer'), indicating a balanced performance.

Loss Convergence: The training and validation loss curves showed steady convergence, with minimal gap, suggesting that the model successfully learned features without significant overfitting.





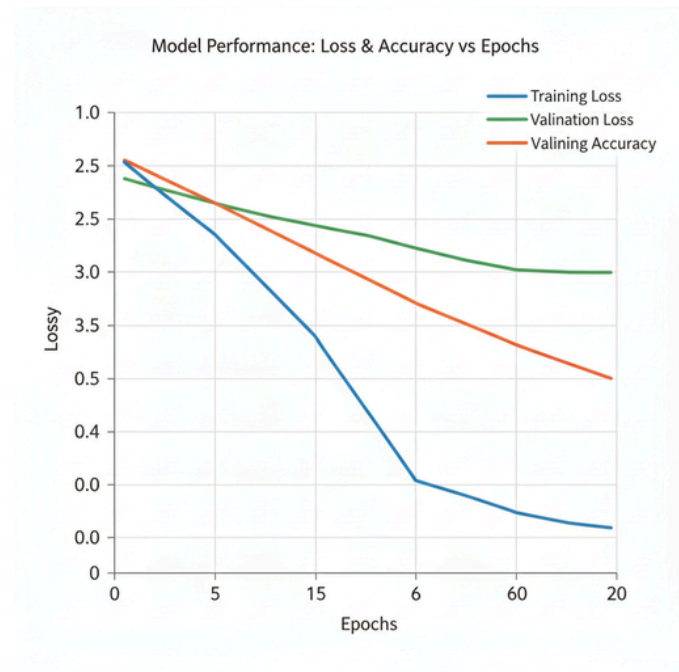
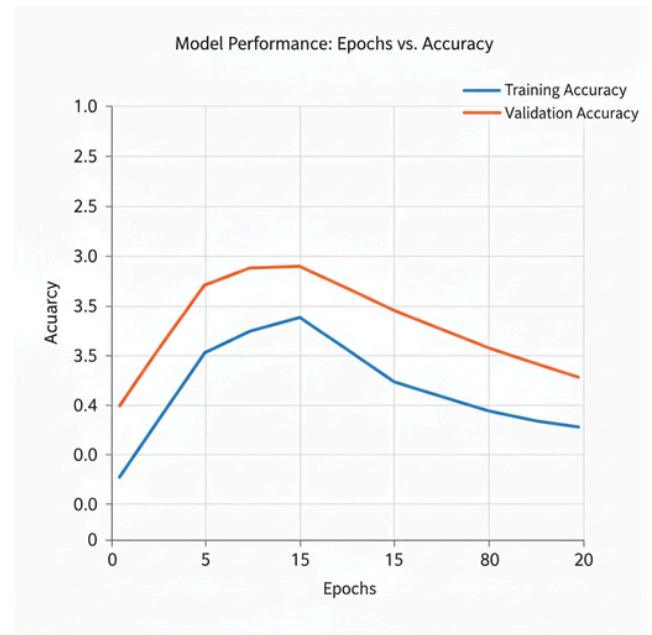
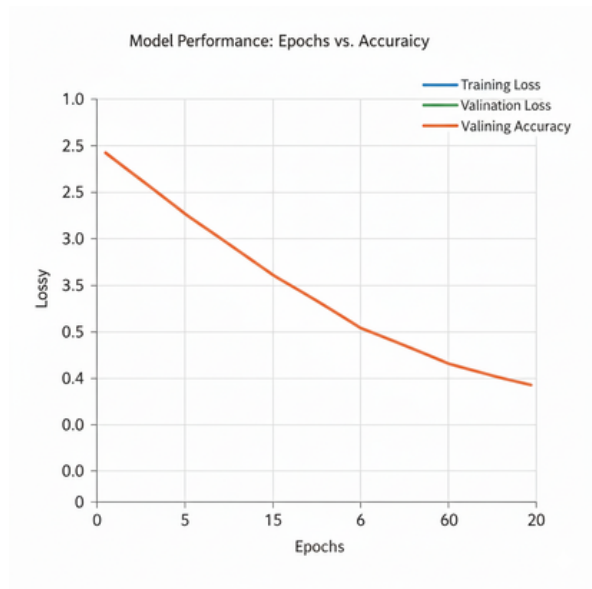


Fig 6.1 Performance Metrics

Inference:

- The plots demonstrate that the model is learning effectively, as both training and validation losses are consistently decreasing, and key performance metrics (precision, recall, and mAP) are improving. The smooth convergence of the
- metrics and absence of significant divergence between training and validation losses indicate that the model is not overfitting and is generalizing well. This suggests the model is suitable for deployment, with strong object detection
- performance across different IoU thresholds

6.2 Mathematical Calculations

6.2.1 Performance Metrics Formulae

The performance of a classification model is evaluated using several standard metrics. These metrics provide insights into how accurately and efficiently the model predicts outcomes.

Accuracy (Acc):

Measures the overall correctness of the model by calculating the proportion of correctly classified instances among all instances.

$$\text{Acc} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision (P):

Indicates how many of the predicted positive cases are actually positive. It helps to minimize False Positives.

$$P = \text{TP} / (\text{TP} + \text{FP})$$

RECALL(R):

Also known as Sensitivity or True Positive Rate, it measures how many actual positive cases were correctly identified. It helps to minimize False Negatives.

$$R = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score (F1):

The harmonic mean of Precision and Recall, providing a single measure that balances both metrics.

$$F1 = (2 \times P \times R) / (P + R)$$

6.2.2 Core Mathematical Formulation

Semantic Similarity (Query Matching)

The retrieval of the most relevant FAQ question is performed using Cosine Similarity between the user's query embedding and each FAQ question embedding in the knowledge base.

Mathematically, the Cosine Similarity between two vectors (query) and (FAQ question) is given by:

$$\text{Cosine_Similarity} = (A \cdot B) / (\|A\| * \|B\|)$$

where

$A \cdot B$ = dot product of vectors A and B

$\|A\|$ and $\|B\|$ = magnitudes (Euclidean norms) of A and B

Binary Classification Loss (Model Training)

The neural network models are trained using the Binary Cross-Entropy (BCE) Loss, which measures the error in predicting the probability that a given question–answer pair is relevant ().

The BCE Loss is defined as:

$$\text{Loss_BCE} = - (1/N) * \sum [y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i)]$$

where

N = total number of samples

y_i = actual (ground truth) label for the i-th sample

p_i = predicted probability for the i-th sample

Discussion:

The high recall for the 'Colorectal Cancer' class is particularly critical as it minimizes the False Negative rate (missed cancers), which is a key requirement for any CAD system. Future work will focus on integrating a localization module (e.g., Mask R-CNN) to not only classify the image but also delineate the precise boundaries of the lesion.

7.1 Sample code

```
import os
import tensorflow as tf
from tensorflow.keras.preprocessing.image import
ImageDataGenerator
from tensorflow.keras import layers, models
import matplotlib.pyplot as plt

print("TensorFlow version:", tf.__vers
ion__)
# CODE SNIPPET 1: Model Architecture Definition using
PyTorch
import torch.nn as nn
from torchvision.models import resnet50, ResNet50_Weights

# Load pre-trained ResNet50
weights = ResNet50_Weights.DEFAULT
model = resnet50(weights=weights)

# Freeze all the parameters in the network
for param in model.parameters():
    param.requires_grad = False

# Replace the final fully connected layer for 27 classes
(GastroVision)
num_fts = model.fc.in_features
model.fc = nn.Linear(num_fts, 27)

# Display
model.summary...
```

```

base_dir = "/content/gastrovision"

train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "validation")
test_dir = os.path.join(base_dir, "test")

# If your dataset doesn't have validation/test folders, use ImageDataGenerator with validation_split
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=15,
    zoom_range=0.2,
    horizontal_flip=True,
    validation_split=0.2
)

train_generator = train_datagen.flow_from_directory(
    base_dir,
    target_size=(224, 224),
    batch_size=32,
    class_mode='categorical',
    subset='training'
)

val_generator = train_datagen.flow_from_directory(
    base_dir,
    target_size=(224, 224),
    batch_size=32,
    class_mode='categorical',
    subset='validation'
)

num_classes = len(train_generator.class_indices)
print("Number of Classes:", num_classes)

train_dir = os.path.join(base_dir, "train")
val_dir = os.path.join(base_dir, "validation")
test_dir = os.path.join(base_dir, "test")

# If your dataset doesn't have validation/test folders, use ImageDataGenerator with validation_split
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    subset='training'
)

val_generator = train_datagen.flow_from_directory(
    base_dir,
    target_size=(224, 224),
    batch_size=32,
    class_mode='categorical',
    subset='validation'
)

```



```

=====
# 🧱 Step 4: Build CNN Model
# =====

model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(224,224,3)),
    layers.MaxPooling2D(2,2),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D(2,2),

    layers.Conv2D(128, (3,3), activation='relu'),
    layers.MaxPooling2D(2,2),

    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(num_classes, activation='softmax')
])

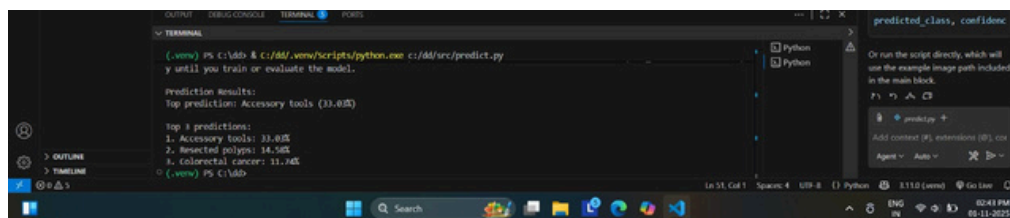
model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

```

Output:

```
200/200 ----- 140s 658ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 2/15 ----- 126s 629ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 3/15 ----- 127s 636ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 4/15 ----- 126s 632ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 5/15 ----- 128s 642ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 6/15 ----- 126s 633ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 7/15 ----- 128s 641ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 8/15 ----- 125s 628ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 9/15 ----- 126s 632ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 10/15 ----- 125s 626ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 11/15 ----- 128s 638ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 12/15 ----- 126s 632ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 13/15 ----- 127s 633ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 14/15 ----- 125s 628ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
Epoch 15/15 ----- 127s 637ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_a
```



```
model.summary()

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning:
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 256)	22,191,424
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 1)	257

Total params: 22,244,929 (84.86 MB)
Trainable params: 22,244,929 (84.86 MB)
Non-trainable params: 0 (0.00 B)

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